Procedure

Make the student to work in pairs to perform the experiment. Follow the steps in given sequence.

1. Put 64 yellow beads in one beaker and 64 green beads in the other to represent male and female gametes respectively. Let the yellow bead be indicated by 'Y' and the green bead by 'y'
2. Take a bead from each container and place them together (it represents fertilization) on the hand towel spread before you on the table.
3. Just like the previous step, continue to pick beads and arrange them in pairs. Thus 64 pairs of beads are obtained representing the 64 heterozygous F_1 progeny.
4. Put 32 F_1 progeny in one petridish and the remaining 32 in another petridish (representing the F_1 males and females).
5. To obtain the F_2 generation, the student should withdraw one bead from one beaker labelled male and one from the other beaker labelled female keeping his / her eyes closed (to ensure randomness) and put them together on the hand towel spread over the table. Continue this process till all the beads are paired. Thus 64 offsprings of F_2 progeny are obtained.
6. Note the genotype (YY or Yy or yy) of each pair and their possible phenotype.
7. Pool all the data and calculate the genotypic and phenotypic ratios.

Observation:

Record the result in the following table:

Generation	Total Number of individuals	Genotypes			Phenotype(s)
		YY	Yy	yy	
F_1					
	Total				
F_2					
	Total				

Phenotypic ratio : in F_1 _____

in F_2 _____

Genotypic ratio : in F_1 _____

in F_2 _____

Inference:

The results are so because when the F_1 individuals are crossed together to raise the F_2 generation, each F_1 individual produces two types of gametes: 50% having dominant allele and the remaining 50% having recessive allele. These gametes undergo random fusion during fertilization to produce the F_2 generation. According to simple probability of mixing of opposite sex gametes, offsprings of three genotypes are likely to appear as follows:

Among these, proportion of dominant phenotype would be YY + Yy = yellow and recessive phenotype yy = green, which occur in 3 : 1 or 75% : 25% ratio.

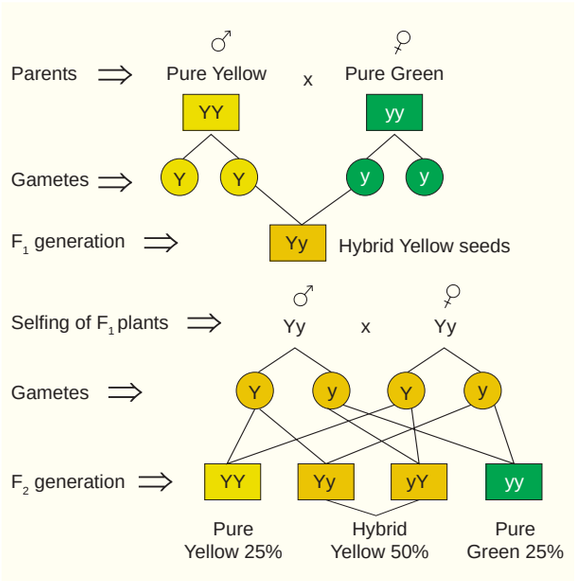


Figure 13 : Monohybrid cross

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